

Sebastian Gomez Ahumada

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Summary

Machine Learning Engineer with hands-on experience shipping production LLM, RAG, and agentic systems on AWS for document-intensive enterprise pipelines. MSc in Biomedical Engineering with research on multimodal data collection, computer vision, and deep learning. Comfortable working end-to-end across ML backend, serverless infrastructure, and full-stack delivery. Focused on turning frontier AI methods — LLM evaluation, retrieval, and human-in-the-loop workflows — into reliable, observable, cost-aware production systems.

Technical skills

Programming: Python, TypeScript/JavaScript, Kotlin, R, SQL, Bash.

ML & AI: PyTorch, TensorFlow, Scikit-learn, LLM fine-tuning, LLM evaluation (LLM-as-judge, DSPy), RAG, agents (Bedrock Converse tool calling, LangGraph, LlamaIndex), Computer Vision, NLP, Signal & image processing.

Retrieval: S3 Vectors, OpenSearch (hybrid BM25 + dense), FAISS.

MLOps & Cloud: AWS (Lambda, Step Functions, SAM, Bedrock, Textract, S3, DynamoDB, ECS Fargate, IAM), Terraform, MLflow, Docker, GitLab CI.

Data & Backend: PostgreSQL, FastAPI, Django, Pandas, NumPy, Pydantic.

Frontend & Mobile: React, Next.js, React Query, Zustand, Vitest, Playwright, Jetpack Compose.

AI-assisted engineering: Claude Code (subagents, worktrees, skills, MCP); ran internal workshops.

Languages: English (Advanced), Spanish (Native).

Education

Universidad de los Andes, MSc in Biomedical engineering. Aug 2023 - Dec 2025

- GPA: 4.74/5
- **Thesis:** Machine Learning Analysis of Colombian Sign Language: Recognition, Classification, and Biomechanical Characterization.
- **Relevant coursework:** Machine learning for engineering, reinforcement learning, analysis and processing of medical images.

Universidad de los Andes, BSc in Biomedical engineering – Minor in Neuroscience. 2019 - 2023

- GPA: 4.08/5
- **Relevant coursework:** Data Structures and Algorithms, Scientific Programming, Signal Processing, Neuroscience, Neuroanatomy.

Experience

Machine Learning Engineer, Provectus – San Francisco, CA (Remote). Jul 2025 – Present

- Built the evaluation system for an LLM contract-extraction platform: an LLM-as-judge evaluator with per-field precision, recall, and F1, tracked in MLflow against golden fixtures and packaged as a reusable library.
- Built most of a production conversational agent over extracted actuarial tables on an accelerated timeline: tool calling through Bedrock Converse (Claude), retrieval over S3 Vectors, and table edits with an audit trail and one-step restore.
- Scaled a Step Functions, Lambda, and ECS Fargate extraction pipeline for very large PDFs, cutting processing time on the largest reports by 52%.
- Agentic document review with LangGraph and hybrid retrieval (OpenSearch) with page-level citations; full-stack delivery with FastAPI, React/TypeScript, Playwright, GitLab CI, and Terraform.

Growth and Software Development Engineer, JustPaid.ai (YC-W23) – San Francisco, CA (Remote, part-time). Jul 2023 – Jun 2025

- Designed and implemented agentic AI workflows for automated information extraction from unstructured

documents (PDFs), including NLP-based text extraction, structured data parsing into Pydantic models, and self-validation loops with agent-based quality supervision.

- Built end-to-end data pipelines integrating ML models with PostgreSQL databases, handling preprocessing, classification, and structured storage of extracted entities (customers, contracts, line items).
- **Languages and tools:** Python, LlamaIndex, LangGraph, Django, PostgreSQL.
- **Contact details:** Daniel Kivatinos - daniel@kivatinos.com

Teaching assistant, Universidad de los Andes – Bogotá, Colombia.

Aug 2023 – Jun 2025

- Conducted laboratory sessions on machine learning fundamentals, covering optimization techniques, linear and logistic regression models, analytical solutions in Ordinary Least Squares (OLS), hyperparameter fine-tuning methodologies, and neural network architectures.
- Managed approximately 70 students per semester. Responsible for laboratory resource creation and student work assessment. **Grade given by students:**4.90. **Languages and tools:** Python, R, Scikit-learn, TensorFlow, PyTorch.
- **Contact details:** Luis Felipe Giraldo Trujillo - lf.giraldo404@uniandes.edu.co

Projects

MSc Thesis: Machine Learning Analysis of Colombian Sign Language: Recognition, Classification, and Biomechanical Characterization.

- I collected data from deaf individuals and Colombian Sign Language (LSC) interpreters using egocentric vision, conventional cameras, and Inertial Measurement Unit (IMU) and Electromyography (EMG) information from a multi-perspective approach in both static and conversational environments.
- I trained machine learning models (Random Forest, K-Nearest Neighbors, and Gradient Boosting) using raw statistical, temporal, and spectral features, and fine-tuned deep learning models (Video-Visual Transformer) from raw videos for sign recognition and deaf-interpreter differentiation. I achieved 95% accuracy in deaf-interpreter classification and 40% accuracy in sign recognition across a dataset of 50 distinct signs.

Developing of a machine learning algorithm for urologic diseases diagnostic support in the "Fundación Santa fé de Bogotá" hospital.

- I developed machine learning models based on decision trees to identify discrepancies in urologic disease diagnostics between doctors and used these models to explain potential reasons for disagreement. We observed 50% diagnostic agreement before standardization sessions, which improved to 75% agreement afterward. Feature importance extracted from the decision trees proved instrumental in enhancing the diagnostic workflow and facilitating consensus among specialists regarding the diagnostic process.

Talks & Publications

- Rios Querubin M, Gomez S. “From Expert Judgment to Autonomous Optimization: Encoding Human Expertise into LLM Judges with DSPy.” *PyCon Colombia 2026* (talk). 2026.pycon.co/talks/33
- Gomez S, et al. “Machine Learning Analysis of Colombian Sign Language: Recognition, Classification, and Biomechanical Characterization.” *Under review*.
- Gomez S, et al. “AI-Based Platform for Automated Uroflowmetry Curve Morphology Classification” and “Improving Uroflowmetry Interpretation: Effects of Standardization Sessions on Interobserver Agreement and AI Model Consistency.” *ICS-EUS 2025* (conference abstracts).

Certificates

Claude Certified Architect - Professional Anthropic.	Sep, 2026
Claude Certified Architect - Foundations Anthropic.	Jun, 2026
AWS Certified Machine Learning Engineer – Associate Amazon Web Services (AWS).	Jan, 2026
Building Transformer-Based Natural Language Processing Applications Nvidia.	Nov, 2023